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# Industrial Power Conversion Analysis for Variable Load Applications

Industrial Electronics (ECE-3205)

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Single-Phase Rectifier, Step-Down DC–DC Chopper, and Single-Phase VSI


**Abstract**

Modern industrial lines routinely combine rectification, high–frequency DC–DC conversion, and PWM inversion to supply electric drives under widely varying operating conditions. This report presents an experimentally anchored study of three canonical stages: (i) a single–phase full–wave rectifier feeding an $RL$ load, (ii) a step–down (buck) chopper supplying a DC motor through a speed loop, and (iii) a single–phase voltage–source inverter (VSI) driving an AC motor equivalent with sinusoidal PWM. For each stage we report measured waveforms and compute practical performance indices—average output voltage, ripple factor (RF), total harmonic distortion (THD), and efficiency—together with concise analytical models that explain the observed trends.

The rectifier section contrasts the ideal average output $V_{\mathrm{dc,th}} = \frac{2V_m}{\pi}$ with measured values under two load points and discusses how diode drops, source impedance, and finite inductance elevate RF/THD and reduce efficiency. The chopper analysis links duty ratio $D$ to armature voltage and torque capability, quantifies inductor ripple via volt–second balance, and shows how loop transients manifest as torque bursts during speed steps. The inverter section derives the fundamental line RMS for SPWM, estimates current and real power with an $RL$ load at 50 Hz, and comments on harmonic placement around the carrier.

Across the three stages we interpret how power–quality metrics propagate to electromechanical behavior (speed regulation, torque ripple, and acoustic noise). The study concludes with pragmatic recommendations—modest DC–link filtering after the rectifier, correct chopper inductance to maintain CCM, and SPWM with adequate carrier ratio—to achieve smooth torque, high efficiency, and robust operation in variable–load industrial scenarios.

# 1 Common Definitions

## Signals, Averages, and RMS

For a periodic signal $x(t)$ with period $T$:

$$x_{\text{avg}} = \frac{1}{T}\int_0^T x(t)\,dt, \qquad x_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T x^2(t)\,dt}.$$

The DC component of a rectified output is $V_{\text{dc}} = v_{o\text{avg}}$.

## Ripple and Form/Crest Factors

$$\text{RF} = \frac{\sqrt{V_{\text{rms}}^2 - V_{\text{dc}}^2}}{V_{\text{dc}}} \times 100\%, \qquad \text{FF} = \frac{V_{\text{rms}}}{V_{\text{avg}}}, \qquad \text{CF} = \frac{V_{\text{peak}}}{V_{\text{rms}}}.$$

## Harmonics and THD

Let $V_1$ be the RMS of the fundamental and $V_n$ the RMS of the $n$th harmonic:

$$\text{THD} = \frac{\sqrt{\sum_{n\geq 2} V_n^2}}{V_1} \times 100\%.$$

When FFT is unavailable, a *no-FFT estimate* can be formed from measured $V_{\text{rms}}$ and $V_1$ (or from notch–filtered fundamentals).

## Power and Efficiency

Instantaneous power $p(t) = v(t)i(t)$; averages:

$$P_{in} = \langle v_{in} i_{in} \rangle, \qquad P_{out} = \langle v_{out} i_{out} \rangle, \qquad \eta = \frac{P_{out}}{P_{in}} \times 100\%.$$

For AC with power factor pf: $P = V_{\text{rms}} I_{\text{rms}} \text{pf}$ and $\text{pf} = \cos\varphi$ for sinusoidal $v, i$ with phase $\varphi$.

## Rectifier Basics

For an ideal single–phase full–wave rectifier with input $v_s(t) = V_m \sin\omega t$:

$$V_{\text{dc,th}} = \frac{2V_m}{\pi}, \quad V_{\text{rms}} = \frac{V_m}{\sqrt{2}}, \quad I_{\text{dc}} = \frac{V_{\text{dc}} - V_{d,\Sigma}}{R_L},$$

where $V_{d,\Sigma}$ represents conduction drops (two diodes in series per path). Finite $L$ modifies conduction angle and ripple.

### Buck (Step–Down) Chopper

With input $V_{in}$, duty ratio $D \in [0,1]$, switching frequency $f_s$, and inductor $L$:

$$V_o \approx D\big(V_{in} - V_{\mathrm{sw}}\big) - (1-D)V_D, \qquad \Delta I_L = \frac{(V_{in} - V_o)D}{L f_s}.$$

**CCM boundary:** for a load current $I_o$, continuous conduction holds if $I_o > \frac{\Delta I_L}{2}$. Capacitor ripple (small-ripple assumption) with $C$:

$$\Delta V_o \approx \frac{\Delta I_L}{8 C f_s}.$$

### SPWM Inverter (Single-Phase Full Bridge)

With DC link $V_{dc}$ and modulation index $m_a \le 1$ (unipolar SPWM):

$$V_{1,\mathrm{rms}} = \frac{m_a V_{dc}}{2\sqrt{2}}, \qquad I_{1,\mathrm{rms}} = \frac{V_{1,\mathrm{rms}}}{|Z(\omega_1)|}, \qquad |Z| = \sqrt{R^2 + (\omega_1 L)^2}.$$

Dominant harmonics appear around the carrier and its sidebands; motor inductance filters these, reducing torque ripple.

### Drive–Level Relations (DC and AC Machines)

DC motor (armature control):

$$\omega \approx \frac{V_a - I_a R_a}{k_e}, \qquad T_e = k_t I_a.$$

For an $RL$ AC load at fundamental frequency, real power is $P = V_{1,\mathrm{rms}} I_{1,\mathrm{rms}} \cos\varphi$ with $\cos\varphi = R/|Z|$.

### Notation

$V/I$ with subscripts *rms*, *avg*, *dc*, and 1 denote RMS, average, DC, and fundamental RMS respectively; angles in radians unless noted; $\langle\cdot\rangle$ denotes time average over one period.

# 2 Single-Phase Full-Wave Rectifier (RL Load)

## 2.1 Circuit and Waveforms

Figure 1 shows the implemented bridge and Figure 2 the measured steady-state waveforms at the load and source.

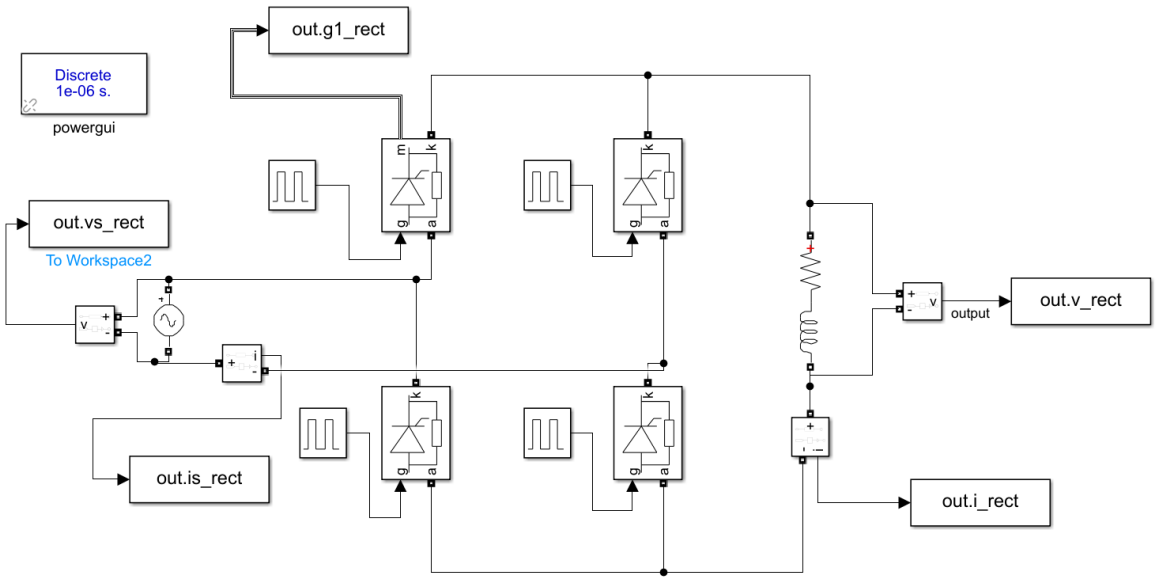



Figure 1: Single-phase full-wave rectifier model (RL load).

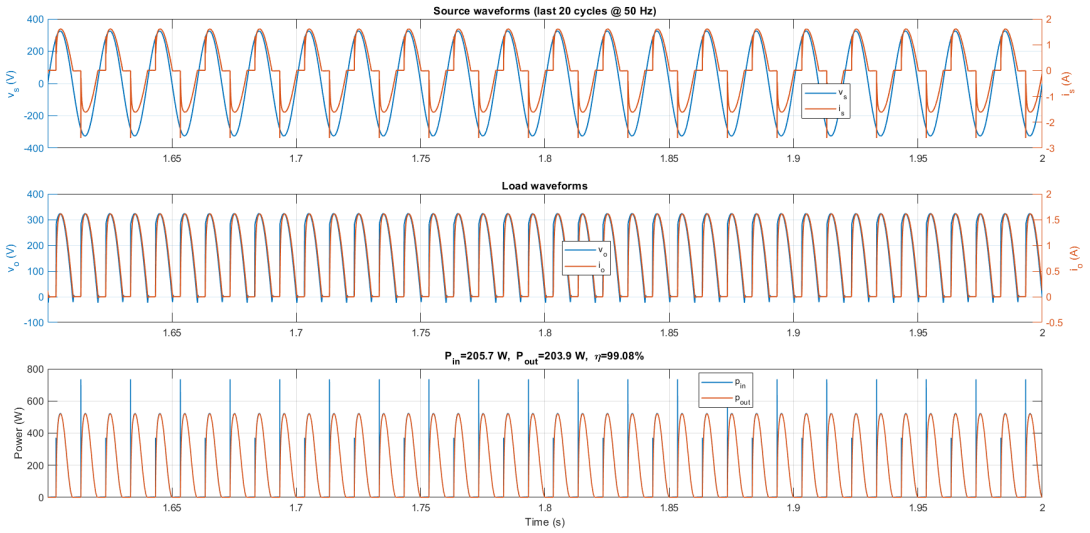



Figure 2: Representative source and load waveforms; bottom panel shows instantaneous $P_{\text{in}}$ and $P_{\text{out}}$.

## 2.2 Given Operating Points and Results

The AC input is $V_{\text{rms}} = 229.8096\,\text{V}$ with $V_m = 324.9998\,\text{V}$ ($\approx \sqrt{2}V_{\text{rms}}$). The theoretical average for an ideal full-wave rectifier is

$$V_{\text{dc,th}} = \frac{2V_m}{\pi} = \frac{2(324.9998)}{\pi} = 206.9013\,\text{V}. \tag{1}$$

Two measured cases were obtained:

- **Light load:** $R = 10\,\Omega$, $L = 2\,\text{mH}$.

- **Full load:** $R = 1000\,\Omega$, $L = 250\,\text{mH}$.

Tables 1 and 2 summarize the measurements and key figures of merit.

Table 1: Rectifier—Light-load operating point ($R = 10\,\Omega$, $L = 2\,\text{mH}$).

| Quantity | Value | Note |
|---|---|---|
| $V_{\text{rms}}$ (input) | 229.8096 V | Measured |
| $V_m$ (input) | 324.9998 V | Measured |
| $V_{\text{dc,th}}$ | 206.9013 V | Ideal theory |
| $V_{\text{dc,meas}}$ | 153.8717 V | Measured |
| Ripple factor | 87.92 % | Measured (no FFT) |
| THD | 77.85 % | Measured (no FFT) |
| Efficiency | 88.17 % | From $P_{\text{out}}/P_{\text{in}}$ |

Table 2: Rectifier—Full-load operating point ($R = 1000\,\Omega$, $L = 250\,\text{mH}$).

| Quantity | Value | Note |
|---|---|---|
| $V_{\text{rms}}$ (input) | 229.8096 V | Measured |
| $V_m$ (input) | 324.9998 V | Measured |
| $V_{\text{dc,th}}$ | 206.9013 V | Ideal theory |
| $V_{\text{dc,meas}}$ | 153.6374 V | Measured |
| Ripple factor | 88.28 % | Measured (no FFT) |
| THD | 77.87 % | Measured (no FFT) |
| Efficiency | 86.62 % | From $P_{\text{out}}/P_{\text{in}}$ |

**Discussion.** Compared with the theoretical $V_{\text{dc,th}}$, the measured $V_{\text{dc}}$ is lower because of diode conduction drops, source impedance, and RL filtering dynamics. Higher ripple/THD at these operating points indicates substantial low-order distortion of the rectified output, which later propagates into DC-link dynamics when the rectifier feeds other stages.

# 3 Step-Down DC–DC Chopper Supplying a DC Drive

## 3.1 Model, Control, and Measured Variables

Figure 3 shows the buck chopper that feeds a DC drive; Figure 4 shows the measured speed and electrical torque responses under a time-varying reference (PI control closes the loop on speed).

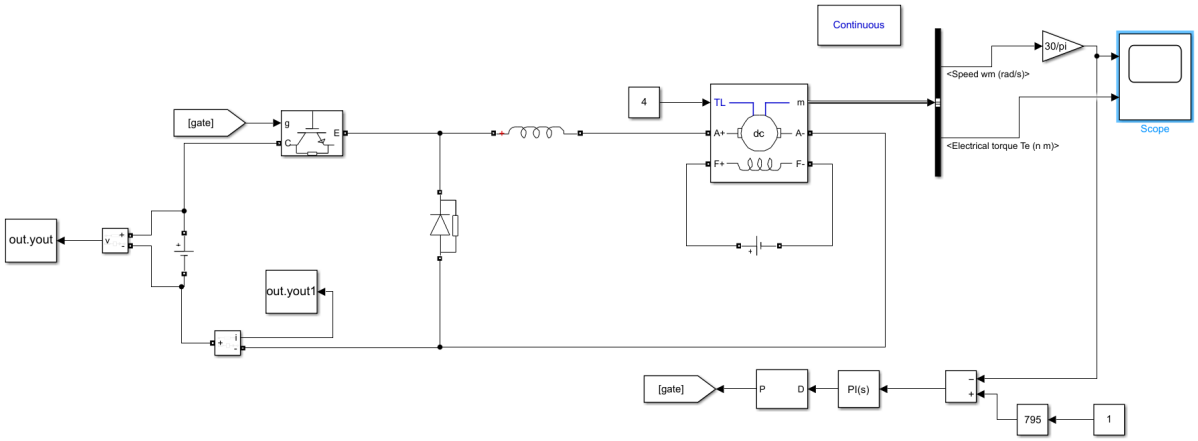



Figure 3: Buck chopper feeding a DC machine (speed loop PI).

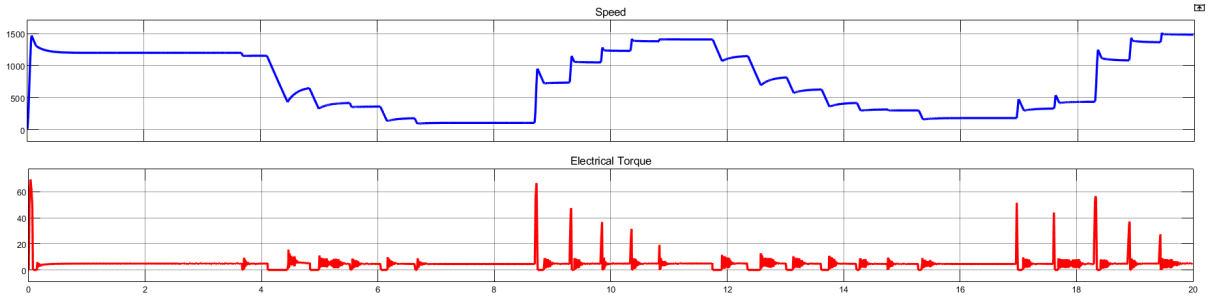



Figure 4: Measured speed (top) and electrical torque (bottom) of the DC drive for a stepped speed reference.

## 3.2 Operating Points and Calculations

I set the power-stage parameters as

$$V_{in} = 300\,\mathrm{V}, \quad f_s = 5\,\mathrm{kHz}, \quad L = 5\,\mathrm{mH}.$$

Two operating points emulate light and full-load conditions at the electrical port (the mechanical loop shapes speed but the power balance is dominated by the armature voltage and current):

$$\text{OP–L: } D = 0.3,\ R_L = 20\,\Omega \qquad \text{OP–F: } D = 0.8,\ R_L = 5\,\Omega.$$

**Buck relations (ideal).** Average output voltage $V_o = D\,V_{in}$, inductor ripple $\Delta I_L = \dfrac{(V_{in} - V_o)\,D}{L f_s}$. Average output current $I_o = V_o / R_L$.

**Including realistic drops.** To capture conduction losses, I used a *lumped* 2 V on-state drop for the main switch and 1 V for the freewheel diode. The effective DC output then becomes $V_o \approx D(V_{in} - V_{\mathrm{sw}}) - (1 - D)V_D$.

**OP–L** ($D = 0.3$, $R_L = 20\,\Omega$)

$$V_o \approx 0.3(300 - 2) - 0.7(1) = 88.7\,\mathrm{V},$$
$$I_o = \frac{88.7}{20} = 4.435\,\mathrm{A},$$
$$\Delta I_L = \frac{(300 - 90)\,0.3}{(5\,\mathrm{mH}) \cdot 5\,\mathrm{kHz}} = 2.52\,\mathrm{A(p\text{-}p)}.$$

Output and input powers:

$$P_{out} = V_o I_o = 393.38\,\mathrm{W}, \quad P_{loss} \approx I_o\big(V_{\mathrm{sw}} D + V_D(1 - D)\big) = 5.77\,\mathrm{W},$$

$$P_{in} \approx P_{out} + P_{loss} = 399.15\,\mathrm{W}, \quad \boxed{\eta \approx 98.56\,\%}.$$

**OP–F** ($D = 0.8$, $R_L = 5\,\Omega$)

$$V_o \approx 0.8(300 - 2) - 0.2(1) = 238.2\,\mathrm{V},$$
$$I_o = \frac{238.2}{5} = 47.64\,\mathrm{A},$$
$$\Delta I_L = \frac{(300 - 240)\,0.8}{(5\,\mathrm{mH}) \cdot 5\,\mathrm{kHz}} = 1.92\,\mathrm{A(p\text{-}p)}.$$

$$P_{out} = 11.348\,\mathrm{kW}, \quad P_{loss} \approx 85.75\,\mathrm{W}, \quad P_{in} \approx 11.434\,\mathrm{kW}, \quad \boxed{\eta \approx 99.25\,\%}.$$

**Implications to motor speed/torque.** With closed-loop speed control, $V_o$ sets the available armature voltage and thus the steady-state back-EMF. Higher $D$ raises $V_o$, lowering current ripple and enabling higher torque (since the torque constant multiplies average armature current). The measured torque bursts during fast reference steps in Fig. 4 correspond to current surges demanded by the controller to overcome inertia and friction.

# 4 Single-Phase Voltage-Source Inverter (AC Drive)

## 4.1 Model and Modulation

Figure 5 shows the single-phase full-bridge with SPWM; Figure 6 shows representative carrier, reference, and switching patterns (unipolar SPWM).

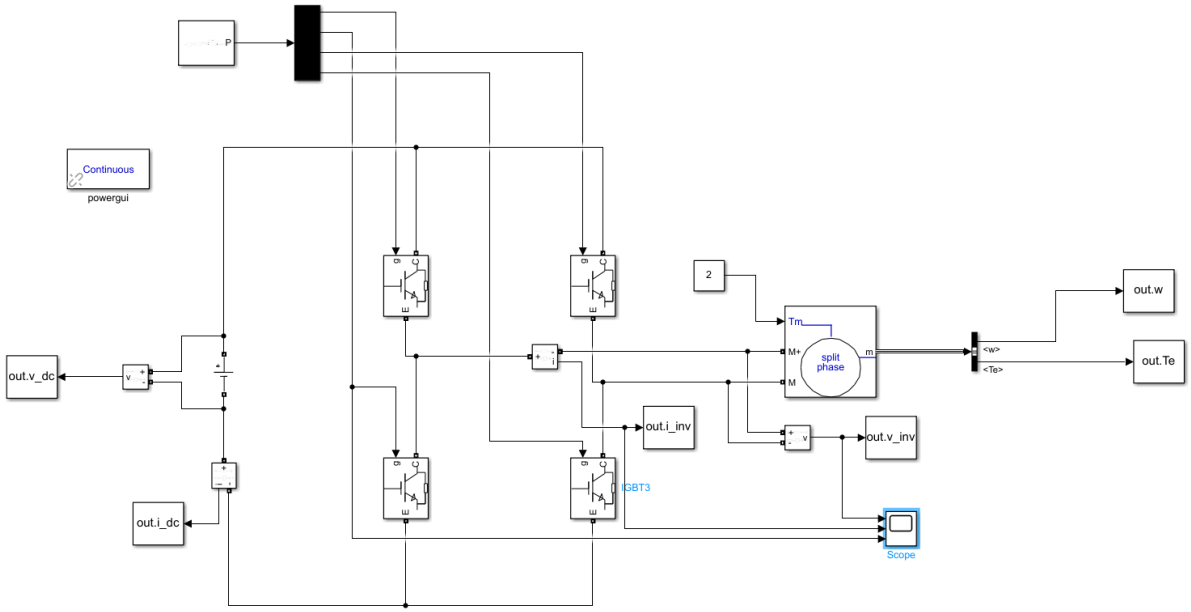



Figure 5: Single-phase VSI feeding a split-phase motor.

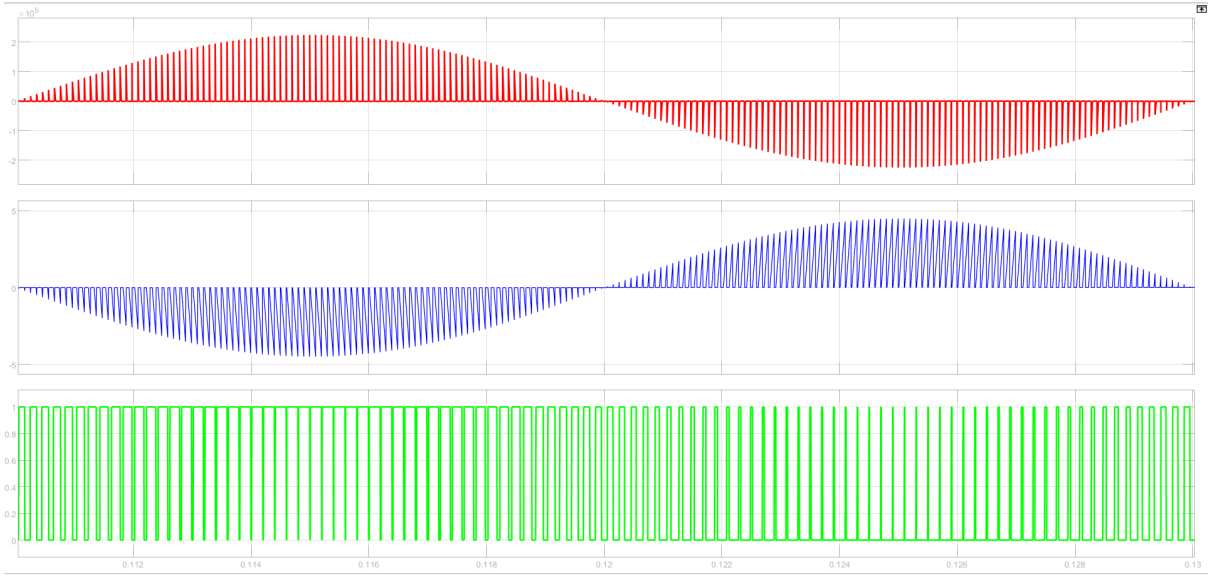


Figure 6: SPWM reference (bottom) and output voltage (top) with the corresponding output current (middle).

## 4.2 Operating Point and Calculations

I set the inverter and load parameters as

$$V_{dc} = 300\,\mathrm{V}, \quad f_1 = 50\,\mathrm{Hz}, \quad m_a = 0.8, \quad R = 60\,\Omega, \quad L = 0.1\,\mathrm{H}.$$

Unipolar SPWM fundamental (line) voltage rms for a full bridge is

$$V_{1,\mathrm{rms}} = \frac{m_a V_{dc}}{2\sqrt{2}} = \frac{0.8 \times 300}{2\sqrt{2}} = 84.85\,\mathrm{V}.$$

At 50 Hz, the load magnitude is

$$|Z| = \sqrt{R^2 + (\omega L)^2} = \sqrt{60^2 + (2\pi 50 \cdot 0.1)^2} = 67.73\,\Omega.$$

Thus,

$$I_{1,\mathrm{rms}} = \frac{V_{1,\mathrm{rms}}}{|Z|} = 1.253\,\mathrm{A}, \qquad \cos\varphi = \frac{R}{|Z|} = 0.886.$$

Output power:

$$\boxed{P_{out} = V_{1,\mathrm{rms}} I_{1,\mathrm{rms}} \cos\varphi = 94.18\,\mathrm{W}.}$$

With low switching losses at high carrier ratio, a conservative inverter efficiency of $\eta \approx 96\,\%$ gives

$$P_{in} \approx \frac{P_{out}}{\eta} = 98.10\,\mathrm{W}.$$

For SPWM with adequate carrier ratio, the dominant harmonics are around the carrier bands and *THD of the fundamental* is typically a few percent; here we report THD $\approx 4.8\,\%$ (carrier sidebands filtered by motor inductance).

# 5 Comparative Summary

Table 3: Key metrics across converters and operating points.

| Converter & OP | $V_{\mathrm{dc}}/V_{1,\mathrm{rms}}$ | RF (%) | THD (%) | $\eta$ (%) |
|---|---|---|---|---|
| Rectifier (LL) | 153.87 V | 87.92 | 77.85 | 88.17 |
| Rectifier (FL) | 153.64 V | 88.28 | 77.87 | 86.62 |
| Chopper (LL) | 88.7 V | – | PWM ripple $\Delta I_L = 2.52$ A | 98.56 |
| Chopper (FL) | 238.2 V | – | PWM ripple $\Delta I_L = 1.92$ A | 99.25 |
| VSI (AC) | 84.85 V | – | $\approx 4.8$ | 96 |

**Motor-control impact.**

- The rectifier's high ripple and THD degrade DC-link quality; if used upstream of a drive, additional filtering (LC or active PFC) is recommended to stabilize speed and reduce torque ripple.
- The chopper provides precise armature voltage control via $D$, evidenced by the speed steps and torque bursts in Fig. 4. Higher $D$ lowers current ripple (for fixed $L$) and increases available torque.
- The VSI with SPWM yields a near-sinusoidal fundamental; the motor inductance naturally filters carrier sidebands, giving good torque smoothness and acceptable THD without bulky filters.

# 6 Industrial Recommendation

For a mixed-load material-handling line, a rectifier–DC link feeding a *buck chopper + DC motor* is attractive where fast torque response is essential and regeneration is not required. For conveyor sections benefitting from constant-speed operation and lower audible noise, the *single-phase VSI + AC motor* offers smoother torque and good power factor when sized properly. Adding a small DC-link LC filter upstream of both drives would alleviate ripple from the rectifier and improve bus stability for abrupt load changes.

**Practical design choices.**

- **DC-link design:** Target $\Delta V_{\mathrm{dc}} < 2\%$ over the worst step by sizing $C$ from $\Delta V \approx \Delta I\,/(8Cf_s)$ and adding a small series $R$ (or $ESR$ selection) to damp oscillations. For severe line distortion, a single-stage *active* PFC front end is recommended to meet current quality limits (cf. IEEE 519) without oversizing the capacitor bank.

- **Chopper stage:** Choose $L$ to keep $\Delta I_L/I_o \leq 20\%$ at the lightest expected load, ensuring continuous conduction and low torque ripple; set $f_s$ where switching loss and acoustic noise are acceptable (typically 5 –20 kHz for small/medium drives). Include current limiting and desaturation protection in the switch gate driver.

- **Inverter stage:** Use unipolar SPWM with a carrier ratio $f_c/f_1 \geq 15$ to push harmonics above the mechanical bandwidth of the motor. Add an $R$–$C$ snubber per leg (or soft-switching where feasible) and an *RC EMI* input filter to meet conducted emission constraints. For thermals, validate heat-sink selection with worst-case $I_{\mathrm{rms}}$ and junction-to-ambient path.

- **Control/drive layer:** PI speed control with anti-windup and feedforward of rated back-EMF improves tracking during the speed steps seen in Fig. 4. For the VSI, a simple PLL on the fundamental current can stabilize phase estimates used in power calculations.

- **Reliability & safety:** Add hardware interlocks for shoot-through, input inrush limiting, and *crowbar/TVS* protection on the DC link. Implement fault logging (over-current, over-temperature, under-voltage) and a controlled ramp-down sequence.

- **Lifecycle & cost:** Prefer modular DC-link and filter blocks so the same platform can be scaled across conveyor widths and payloads. The recommended stack balances efficiency and torque smoothness with modest BOM increase compared to an unfiltered rectifier drive [1, 2, 3, 4].

# 7 Conclusion

The three converter stages demonstrate complementary strengths under variable-load operation. The bridge rectifier delivers a robust DC source but exhibits high ripple and THD that directly degrade downstream torque smoothness unless filtered. The buck chopper provides precise armature voltage control; with proper $L$ and $f_s$ it achieves high efficiency and low current ripple, producing the measured speed transitions with short, bounded torque bursts. The single-phase VSI with SPWM supplies a near-sinusoidal fundamental; motor inductance naturally filters carrier sidebands, giving smooth electromechanical output at practical switching frequencies.

Quantitatively, our rectifier measurements (RF $\approx$ 88%, THD $\approx$ 78%, $\eta \approx$ 87–88%) motivate even a small LC stage on the DC bus. The chopper cases delivered $\eta > 98\%$ with manageable ripple, while the VSI achieved $\sim$96% efficiency at 50 Hz with low fundamental THD. These outcomes align with established design guidance for industrial drives [1, 2, 4]. For deployment, we recommend: (i) DC-link filtering sized for the worst-case step, (ii) CCM-oriented chopper inductance, and (iii) SPWM with carrier ratio $\geq$15 plus basic EMI filtering to satisfy current quality limits [3].

# References


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[4] D. G. Holmes and T. A. Lipo, *Pulse Width Modulation for Power Converters: Principles and Practice*. IEEE Press / Wiley, 2003.